Efficient Semantic Splatting for Remote Sensing Multiview Segmentation

Zipeng Qi, Hao Chen, Haotian Zhang, Zhengxia Zou, *Senior Member, IEEE*, and Zhenwei Shi, *Senior Member, IEEE*

***Abstract*— Remote sensing multiview image segmentation is essential for achieving accurate and consistent stereoscopic perception of target scenes. This task involves processing RGB images from multiple viewpoints to generate high-accuracy, view-consistent semantic segmentation across all views. Traditional training-based methods struggle with maintaining cross-view consistency, while optimization-driven approaches using implicit neural networks improve view consistency but suffer from slow parameter optimization and inference. To overcome these limitations, we propose a novel Gaussian splatting-based semantic segmentation framework. Our method efficiently projects the color attributes and semantic features of 3-D Gaussians onto the image plane, enabling the simultaneous generation of both RGB images and segmentation outputs. By leveraging explicit spatial structures and a splatting rendering strategy, our approach significantly enhances optimization efficiency and rendering speed. In addition, we incorporate SAM2 to generate pseudo-labels for boundary regions, addressing the lack of supervision in sparsely labeled views (e.g., 3%). To further enforce cross-view consistency and feature coherence of 3-D Gaussians, we introduce a two-level aggregation loss that operates at both the 2-D feature map and 3-D spatial levels. Extensive experiments across nine datasets demonstrate the superiority of our method, achieving competitive segmentation quality with limited supervisory views. Notably, our approach reduces rendering (inference) times by 90%, while improving the average mean intersection over union (mIoU) by up to 3.5%.**

***Index Terms*— Gaussian splatting, remote sensing, SAM2, semantic segmentation.**

I. Introduction

REMOTE sensing image segmentation is a fundamental technology widely applied in various tasks, including change detection [1], [2], [3], [4], object extraction, e.g., (road and building) [5], [6], [7], and scene understanding [8], [9], [10], [11]. While these tasks typically produce single-view results, this article focuses on the multiview image segmentation task for remote sensing images, aiming to provide a more comprehensive understanding of target scenes. The inputs are RGB images captured from hundreds of views of a scene, with the outputs being segmentation results for each view. Due to the labor-intensive nature of semantic labeling, we aim to achieve multiview segmentation using a small, easily obtainable set of annotated images—for instance, using only 3% of labeled images from the target scene. With multiview segmentation, we can effortlessly construct downstream outputs, such as mesh or point cloud segmentation.

Traditional segmentation networks, including CNN-based [12], [13], [14], [15] and Transformer-based [16], [17], [18], [19], [20] approaches, can be directly applied to this task. However, these methods often rely on densely labeled data and extensive training resources. Moreover, they process each view independently, making it challenging to capture relationships between views or reconstruct the 3-D structure of the scene, resulting in poor view consistency and low accuracy, as shown in Fig. 1. While some approaches leverage pretrained or fine-tuned depth estimation modules to incorporate spatial information, these methods often demand even greater resources and struggle with out-of-distribution (OOD) data. Therefore, achieving multiview segmentation with robust view consistency under sparse labeling conditions is essential to address these challenges while minimizing the reliance on exhaustive annotations and computational resources.

Recently, optimization-based segmentation methods [21], [22], [23] have shown the ability to generate multiview segmentation results using limited semantic labels. These methods divide the space covered by the views and often utilize implicit neural networks, such as NeRFs [24], [25], [26], to map spatial positions to attributes like color and semantic information for each sampled point. Through volume rendering techniques, they can generate both color and segmentation outputs while optimizing these attributes using self-supervision. This approach implicitly constructs a continued spatial representation of the scene, thereby improving view consistency. However, the optimization process and the

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representation structure of implicit neural networks are often inefficient, resulting in significant optimization and rendering latencies, as illustrated in Fig. 1.

In general, training-based methods primarily focus on single-view images and require a large number of labeled data, limiting their capacity to model relationships between different views and resulting in poor performance when labels are sparse. On the other hand, optimization-based methods, while capable of achieving better view consistency, often suffer from high latency during both optimization and rendering due to their reliance on implicit representations. To overcome these challenges, we propose an efficient (typically saving 90% testing time assumption) Gaussian splatting-based semantic splatting method tailored for remote sensing scenes. Our approach is designed to achieve high performance (increasing average accuracy over 3.5%) with a minimal number of labeled views (e.g., three labeled images per 100 views).

Unlike continuous color-related attributes that can be directly optimized, semantic classes are discrete and require a different approach. To address this, we first incorporate additional continuous semantic features for each point and project these features onto the image plane to generate a semantic feature map. Subsequently, a transfer layer is introduced to derive the final segmentation results from the projected feature map. Second, the wide shooting angles in remote sensing scenes pose a challenge for accurately modeling semantic attributes, particularly in boundary regions, due to the limited number of labeled views. To overcome this, we utilize foundation models such as SAM2 [27], which are trained on extensive datasets, to generate pseudo-labels for boundary areas. This approach improves segmentation accuracy in these challenging regions. Finally, while the basic Gaussian splatting [28] architecture enhances computational efficiency, it may lead to a loss of spatial continuity in attribute modeling. To address this issue, we design two aggregation losses to enforce spatial consistency at both the 2-D feature map level and the 3-D spatial point level, ensuring more coherent attribute representation.

We conduct extensive experiments to validate the effectiveness of the proposed method. The evaluation is performed on a multiview segmentation dataset, which includes six sets of synthetic data generated using the well-known CARLA simulation platform [29] and three sets of real-world data collected from Google Maps. Experimental results demonstrate that our method outperforms both training-based and optimization-based state-of-the-art approaches. Furthermore, visual comparisons highlight the ability of our method to deliver more accurate and view-consistent segmentation results.

The contributions of this article are summarized as follows.

1) We are the first to extend Gaussian splatting to remote sensing multiview image semantic segmentation. Our proposed method incorporates a semantic vector for each sampled point and a transfer layer to generate segmentation results. Experimental results demonstrate that our approach outperforms NeRF-based methods in both accuracy and efficiency.

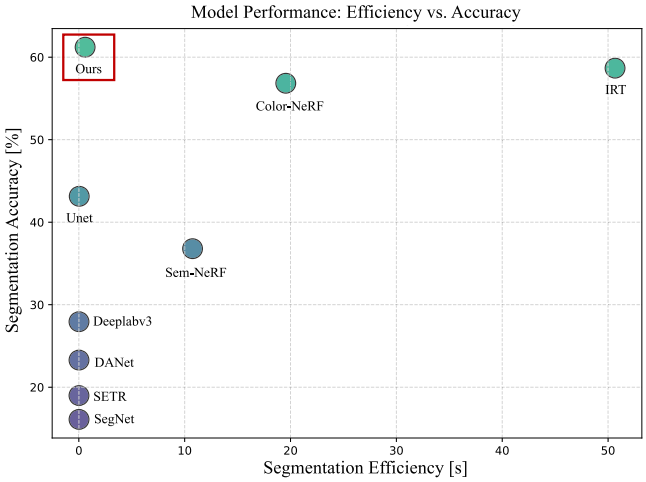


Fig. 1. Our method achieves superior efficiency and accuracy for multiview segmentation using limited labels. Training-based methods: SegNet, SETR, DeepLab, and Unet. Optimization-based methods: Sem-NeRF, Color-NeRF, IRT, and Ours. The testing time of our method is comparable with CNN-based and Transformer-based methods.

2) Limited supervision across multiple views often leads to weak boundary supervision, resulting in inaccurate segmentation. To address this challenge, we transfer the knowledge of SAM to 3-D scenes, effectively improving the supervision quality in boundary regions and enhancing segmentation accuracy.
3) To enhance the spatial consistency of semantic attributes for 3-D Gaussians, we design a two-level aggregation loss that enhances spatial consistency at both the 2-D feature map level and the 3-D spatial point level. This improves the spatial continuity of semantic features and overall segmentation performance.

The remainder of this article is organized as follows. Section II reviews the related work. Section III details the proposed method. Experimental results are presented in Section IV. Finally, the conclusion is provided in Section V.

II. Related Work

To provide a clearer understanding of our method and its distinctions from existing approaches, we review the related work and preliminary of coordinate system conversion as follows.

A. Training-Based Segmentation

Training-based methods have benefited significantly from the rapid advancements in convolutional neural networks (CNNs) and Transformer architectures, achieving remarkable success in various segmentation tasks. CNN-based models primarily address challenges such as preserving detailed information and extracting accurate and rich abstract features. Many of these models are built on an encoder–decoder framework, with Unet [30] being a prominent example. In this architecture, the encoder extracts image features through cascaded convolutional kernels while progressively downsampling the feature map resolution. The symmetrical decoder then reconstructs the feature maps and upsamples

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their resolution. Skip connections between the encoder and decoder are employed to retain fine-grained details during the transformation from RGB images to semantic labels. Building on this framework, dilated convolution [31] has been introduced to expand the receptive field without sacrificing resolution. This technique reduces network parameters and mitigates the risk of over-fitting. In addition, some approaches incorporate multiscale information fusion and feature pyramid strategies [32], [33], [34], [35] to capture richer abstract features and improve segmentation accuracy. Methods such as DANet [36] further enhance feature representation by leveraging dual attention mechanisms, effectively capturing both coarse object-level and fine pixel-level information to enrich feature extraction.

Recently, the Transformer architecture, initially introduced in the field of natural language processing (NLP) [37], has gained widespread adoption in the image processing domain for tasks such as classification, object detection, and segmentation. This architecture treats image patches as tokens and processes them through cascade layers, where the attention mechanism effectively handles long-range dependencies and explores relationships between tokens. SETR [16] was the first to extend the Vision Transformer [38] framework to image segmentation tasks. Building on this, Segformer [39] utilizes a lightweight backbone combined with simplified MLP-like prediction heads, achieving higher accuracy while maintaining efficiency. Despite these advancements, training-based methods typically rely on large volumes of pixel-dense annotations to learn robust and generalizable features, ensuring high accuracy. However, they often struggle to model spatial relationships across multiple views, limiting their effectiveness in multiview scenarios.

Some methods [40], [41] combine RGB and depth information by utilizing a dual-branch CNN backbone or incorporating self-attention mechanisms to address the limitations in modeling spatial relationships. However, these approaches rely on additional depth supervision for parameter training, which is often expensive and impractical to obtain in real-world applications. Furthermore, pretrained depth estimators typically struggle with multiview depth prediction, as they are usually trained on single-view datasets and fail to generalize effectively to OOD data. In contrast, our method achieves the desired results using only RGB images and labels from a limited number of views, eliminating the need for additional depth supervision.

B. Optimization-Based Segmentation

Optimization-based methods leverage implicit neural networks, such as NeRF, which were initially designed for novel view synthesis, to process single scenes captured from multiple views. NeRF-like approaches [21], [22], [23], [42], [43], [44] divide 3-D space by sampling spatial points along rays extending from the camera to each pixel. These methods assign color and density attributes to each sampled point using multilayer perceptions (MLPs). The MLPs take the position and view angle of each point as input and output the corresponding attributes, as follows:

$$c_i, \sigma_i = \Phi_d((\Phi_s(x, y, z), \theta, \beta)) \tag{1}$$

where x, y, z, θ, and β are the ith point position and the angle of the associated ray, the c_i and σ_i are the color and density values corresponding to the ith point, respectively, while Φ_d and Φ_s are the MLPs. Then, they render the final color values of each pixel with volume rendering as follows:

$$\text{Color} = \sum_{i=1}^{N} \exp\left(-\sum_{j=1}^{i-1} \alpha_j \sigma_j\right)(1 - \exp(-\alpha_i \sigma_i))c_i \tag{2}$$

where α_i is the interval distance between the i point and the $i+1$ point along the corresponding ray. They compare the rendered results and captured images of each view within a self-supervision to optimize the parameters of MLPs. The optimized MLPs establish a query relationship between positions and attributes. Accurate query mapping enables effective rendering of images from novel views.

Several methods extend the rendering process from RGB images to semantic labels. For instance, Sem-NeRF [22] introduces an additional branch to generate semantic features for each sampled point, rendering both semantic labels and RGB images using a volume rendering approach. Panoptic-NeRF [21], [45] further improves segmentation performance in scenes with minimal viewpoint variation by incorporating coarse 3-D bounding boxes and noisy 2-D predictions as supervisory signals. Other methods, such as NeSF [46], recover 3-D density fields and train a 3-D semantic segmentation model, supervised by posed 2-D semantic maps, though this approach requires higher computational resources. Similarly, some methods [44], [47], [48] rely solely on positional information to determine semantic features. Furthermore, Color-NeRF [42] enhances the segmentation of small objects by incorporating color information into the rendering process for semantic features. Similarly, RFP [49] proposes a joint encoding strategy that embeds semantic information, appearance features, and geometric attributes into the neural radiance field. In addition, large pitch angles can lead to a significant number of sampling points in the air, which negatively impacts rendering accuracy. To mitigate this issue, IRT [23] introduces a selection layer to optimize the sampling and rendering processes and enriches the integration of color information through a lightweight texture extraction network. Recently, some works [50], [51], [52] have combined NeRF-based segmentation with the SAM model for unsupervised segmentation. However, these methods primarily focus on formulating object segmentation rather than outputting detailed category information, which is different from our task.

It is also noteworthy that NeRF-like methods naturally support the extraction of depth maps or mesh representations using density attributes. Compared to traditional approaches such as TSDF [53] or point clouds [54], implicit spatial representations offer distinct advantages, including a smaller memory footprint and spatial resolution independence.

With the development of Gaussian splatting, several methods [55], [56], [57] have extended its application

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from RGB rendering to segmentation tasks. However, these approaches predominantly focus on indoor scenes and rely on all-view supervision to achieve instance identification. In contrast, our work targets remote sensing scenes and emphasizes semantic segmentation, particularly under sparse-view supervision. The basic preliminary of Gaussian splatting will be introduced in Section III-A.

Since the NeRF-based architecture implicitly models spatial point positions and their corresponding attributes within a neural network, it exhibits a certain degree of robustness to variations in the attributes of adjacent points. As a result, multiview segmentation methods based on NeRF tend to produce more consistent results across different views. In contrast, the 3-D Gaussian multiview segmentation method employs an explicit point cloud-based representation of spatial information, which leads to lower continuity compared to the implicit NeRF approach. So these are many works [55], [56] that introduce regularization losses to enforce semantic feature similarity among neighboring spatial points. However, due to its explicit representation, 3-D Gaussian achieves significantly faster rendering speeds than NeRF-based methods.

C. Segment Anything

Segment anything (SAM) [58] is a foundational vision model designed to generate accurate image segmentation based on input visual prompts, such as points or bounding boxes. The model consists of three key modules: an image encoder, a prompt encoder, and a mask decoder. The masks produced by SAM represent regions that may contain the target objects. Depending on the number and positioning of the prompts, SAM generates potential segmentation results at multiple levels. Trained on a large volume of data, SAM demonstrates robust performance and excels across a wide range of segmentation tasks [51], [59], [60], [61]. SAM2 [27] builds upon SAM, extending its capabilities from the image domain to the video domain. SAM2 can accept various types of prompts, including clicks (positive or negative), bounding boxes, or masks, to define the extent of an object within a specific video frame. The lightweight mask decoder then outputs a segmentation mask for the current frame based on the image embedding and encoded prompts. In the video setting, SAM2 propagates the mask prediction across all frames. Additional prompts can be iteratively provided in subsequent frames to further refine the segmentation. To ensure accurate mask predictions across all frames, SAM2 incorporates a memory mechanism, consisting of a memory encoder, memory bank, and memory attention module.

However, both SAM and SAM2 are limited to generating nonsemantic masks, which are not directly applicable to remote sensing images. Fortunately, these nonsemantic masks are adequate for our approach. We use SAM2 to generate pseudo-labels for boundary regions to improve the segmentation results.

D. Coordinate System Conversion

Our method involves the conversion of four coordinate systems including the pixel coordinate system, image coordinate system, camera coordinate system, and world coordinate system. Therefore, in this section, we introduce transformations between different coordinate systems.

The conversion relationship between the pixel coordinate system and the image coordinate system is defined as follows:

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{d_x} & 0 & u_0 \\ 0 & \frac{1}{d_y} & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \tag{3}$$

where x, y and u, v represent the pixel and image coordinates, respectively, d_x and d_y are the width and height of a pixel corresponding to the photosensitive point. The camera coordinate system to the image coordinate system follows a perspective transformation relationship, which can be represented by using similar triangles:

$$z_c \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = [\mathbf{K}|\mathbf{0}] \begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} \tag{4}$$

where x_c, y_c, and z_c are the camera coordinate, f is the focal length and K is the intrinsic parameter matrix of the camera. The world coordinate system can be obtained from the camera coordinate system through rotation and translation

$$\begin{bmatrix} x_c \\ y_c \\ z_c \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}_{1\times 3} & 1 \end{bmatrix} \begin{bmatrix} x_w \\ y_w \\ z_w \\ 1 \end{bmatrix} \tag{5}$$

where R represents the rotation matrix and t the translation matrix, with $[R|t]$ being the camera's extrinsic parameters. The coordinates x_w, y_w, and z_w refer to the absolute positions of objects in the world coordinate system. In our work, we employ COLMAP [62] to estimate the intrinsic and extrinsic parameters for each viewpoint.

III. Methodology

An overview of our method is illustrated in Fig. 2. We propose an efficient, optimization-based semantic splatting approach for multiview segmentation in remote sensing. The input to our method consists of a set of RGB images, $\mathbf{I_{1:N}}$, captured from various views, the internal and external parameters of each view/camera, $\mathbf{V}_{1:N}$, along with a limited number of labels, $\mathbf{L_{1:M}}$, where $M \ll N$. The output is the semantic segmentation results for the testing views or any novel views. Our approach leverages Gaussian splatting [63], which utilizes an explicit point cloud structure. For each point in the cloud, we assign a Gaussian, $\mathbf{G}$, with both spatial, color, and semantic feature attributes.

We begin by initializing the point cloud using COLMAP and setting the parameters for each Gaussian. Then, we randomly select a view and render both color and segmentation results. After calculating the loss, the parameters are updated accordingly. This process of rendering and parameter updating is repeated iteratively. The details of the rendering process can be found in Sections III-A and III-B. The explicit point

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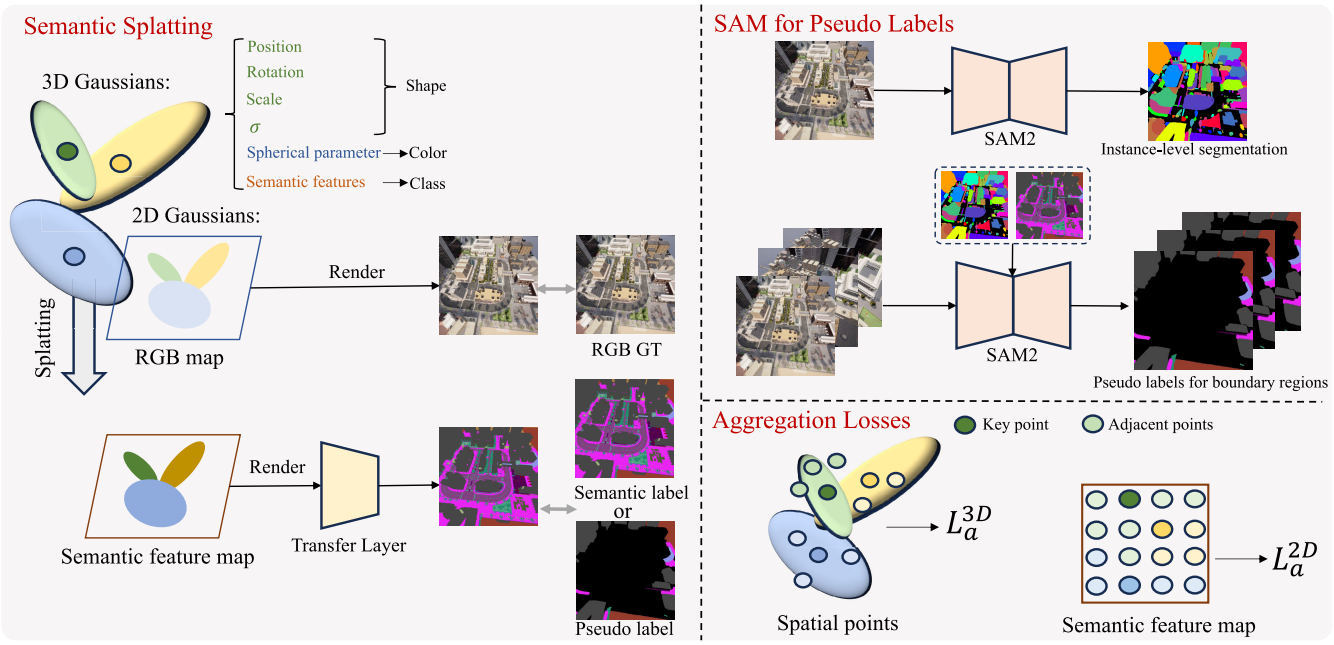


Fig. 2. Overview of the proposed model. Our method is an optimization-based semantic splatting approach for multiview segmentation in remote sensing. Leveraging an explicit point cloud structure and volume rendering, it achieves high accuracy and low latency in generating RGB and semantic segmentation results (see Sections III-A and III-B). Aggregation losses improve spatial generalization (see Section III-D), while SAM2 generates pseudo-labels for boundary regions, further enhancing boundary segmentation quality (see Section III-C).

cloud structure enhances the efficiency of optimization and rendering. However, it exhibits weaker spatial generalization capabilities compared to implicit neural networks, particularly in boundary regions under sparse views with supervision. To address this limitation, we introduce aggregation losses at both the 2-D feature map and 3-D spatial point levels (see Section III-D). In addition, we employ SAM2 to generate pseudo-labels for the boundary regions of each view (see Section III-C). The detailed optimization and inference process of our method is provided in Algorithm 1.

Next, we provide a detailed description of the splatting process, noting that color splatting and semantic splatting follow similar procedures. For the selected view, both color and segmentation can be rendered simultaneously using color and semantic splatting.

A. Color Rendering

In this section, we introduce splatting-based rendering, the core of our color and semantic rendering approach. Unlike ray casting, splatting iteratively projects and accumulates spatial positions using a footprint function to determine their influence range. By modeling intensity distributions with Gaussian functions, it efficiently synthesizes contributions to generate the final image. Its low-latency nature makes splatting widely adopted in graphics applications [64], [65], [66]. Gaussian splatting [28] extends splatting technology to the field of novel view synthesis.

Building on Gaussian splatting, our color rendering process begins by using COLMAP to initialize a point cloud from multiview RGB images. Each point in the cloud is then assigned a 3-D Gaussian, serving as the foundation for subsequent rendering.

The spatial properties of each 3-D Gaussian are defined by its 3-D mean, $\mu^{3\text{-D}} \in \mathbb{R}^3$, represented as (x, y, z), which corresponds to the position of the associated spatial point, and its 3-D covariance matrix, $\Sigma^{3\text{-D}} = RSS^TR^T$, which characterizes the radiance field. Here, $S \in \mathbb{R}^3$ and $R \in \mathbb{R}^{3\times3}$ denote the scaling and rotation components of the Gaussian ellipsoid, respectively. The density attribute $\sigma \in \mathbb{R}^1$ represents opacity. For color representation, we use a set of spherical harmonics coefficients, $H \in \mathbb{R}^{27}$. Following [28], the color vector $c \in \mathbb{R}^3$ of the 3-D Gaussian is computed based on the rendering angle determined by the camera parameters and H, capturing the view-dependent appearance of the scene.

During the color splatting process, 3-D Gaussians are first projected onto the image plane corresponding to the selected view as 2-D Gaussians through a coordinate system transformation, defining their 2-D influence regions. Specifically, the 2-D mean $\mu^{2\text{-D}} \in \mathbb{R}^2$, representing the pixel position, is derived by transforming the 3-D mean $\mu^{3\text{-D}}$ using the camera parameters. This process is defined in the Section II-D. The corresponding 2-D covariance matrix $\Sigma_i^{2\text{-D}} \in \mathbb{R}^{2\times2}$ is computed as

$$\Sigma_i^{2\text{-D}} = JW\Sigma_i^{3\text{-D}}W^TJ^T \tag{6}$$

where J is the Jacobian of the affine approximation of the 3-D–2-D projection, and W is the world-to-camera transformation matrix.

Next, for a given pixel position $p \in \mathbb{R}^2$, we determine the influencing 2-D Gaussians using $\mu^{2\text{-D}}$ and $\Sigma^{2\text{-D}}$, and then sort them based on their distance to the image plane, resulting in n valid 2-D Gaussians. Subsequently, alpha blending—a technique for mixed alpha synthesis—is applied to calculate

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Algorithm 1 Semantic Splatting

Input: $\{\mathbf{I}_{1:N}, \mathbf{V}_{1:N}, \mathbf{L}_{1:M}\}$ // N images $\mathbf{I}$ with camera parameters $\mathbf{V}$ and M labels $\mathbf{L}$, where $M \ll N$
Input: $\{G_1, G_2, \ldots, G_k\}$ // k initialized 3D Gaussians
Input: $\{\mathbf{T}, \mathbf{P}_{1:N}, \mathbf{B}_{1:N}\}$ // Iteration time, Pseudo labels, and boundary masks
Output: Optimized 3D Gaussians

// Optimization Process
for $i \leftarrow 1$ ***to*** $\mathbf{T}$ **do**
  // Select an view and obtain corresponding image and camera parameters
  $\mathbf{I} \leftarrow \text{Sample}(\mathbf{I}_1, \mathbf{I}_2, \ldots, \mathbf{I}_N)$
  $\mathbf{V} \leftarrow \text{Sample}(\mathbf{V}_1, \mathbf{V}_2, \ldots, \mathbf{V}_N)$
  $C^{\text{all}}, S_f^{\text{all}} \leftarrow \text{Splatting}(G_1, G_2, \ldots, G_k \mid \mathbf{V})$ // Eq. (7) and Eq. (9)
  $S^{\text{all}} \leftarrow \text{Decoder}(S_f^{\text{all}})$ // Eq. (10)
  if $\mathbf{I}$ *has supervision* **then**
    $\mathbf{L} \leftarrow \text{Loss}(\{C^{\text{all}}, \mathbf{I}\}, \{S^{\text{all}}, \mathbf{L}\})$
  else
    $\mathbf{L} \leftarrow \text{Loss}(\{C^{\text{all}}, \mathbf{I}\}, \{S^{\text{all}}, \mathbf{P}\}, \mathbf{B})$
  end
  // Calculate the loss using Eq. (14)
  Update parameters in 3-D Gaussians
end

// Inference Process
// Select a test view and obtain corresponding camera parameters
$\mathbf{V} \leftarrow \text{Select}(\mathbf{V}_1, \mathbf{V}_2, \ldots, \mathbf{V}_N)$
$_, S_f^{\text{all}} \leftarrow \text{Splatting}(G_1, G_2, \ldots, G_k \mid \mathbf{V})$ // Eq. (9)
$S^{\text{all}} \leftarrow \text{Decoder}(S_f^{\text{all}})$ // Eq. (10)

the final color as follows:

$$C^{\text{one}} = \sum_{i \in n} c_i \alpha_i' \prod_{j=1}^{i-1} (1 - \alpha_j') \tag{7}$$

where C^{one} is the render color vector, c_i is the color attribute vector of the ith Gaussian, and the α_i' is the opacity of the ith Gaussian splatted in the pixel position p

$$\alpha_i' = \sigma_i \exp\left(-\frac{1}{2}\left(p - \mu_i^{2\text{-D}}\right)^T \left(\Sigma_i^{2\text{-D}}\right)^{-1} \left(p - \mu_i^{2\text{-D}}\right)\right). \tag{8}$$

For other pixels, we apply the same method and batch the processing of all points, enabling parallel processing of all pixels to obtain the final result. Typically, the $\mathcal{L}_1$ loss and $\mathcal{L}_{D-\text{SSIM}}$ loss [28] are employed to compute the loss, which is then used to optimize all relevant parameters in a self-supervised manner. The accuracy of color rendering results plays a crucial role in scene spatial structure estimation, which also serves as vital information for semantic splatting.

B. Semantic Rendering

In this section, we present the process of semantic rendering. Unlike color attributes, which exist in a continuous domain, semantic classes are discrete values, making the direct optimization of semantic classes for each point a nondifferentiable problem. To address this, we transform the nondifferentiable optimization into a differentiable process. Specifically, we elevate the 1-D discrete class values into high-dimensional continuous semantic features $s \in \mathbb{R}^{16}$, which are then assigned to each 3-D Gaussian, as shown in Fig. 2. The 3-D projection process follows the same procedure as in color splatting. We render the semantic feature $S_f^{\text{one}} \in \mathbb{R}^{16}$ in the pixel position p as follows:

$$S_f^{\text{one}} = \sum_{i \in n} s_i \alpha_i' \prod_{j=1}^{i-1} (1 - \alpha_j'). \tag{9}$$

The spatial information including $\mu^{3\text{-D}}$, $\Sigma^{3\text{-D}}$, and α as well as the splatting process, is shared between color and semantic rendering. Here, s_i represents the semantic attribute of the ith 3-D Gaussian. For other pixels, we apply the same method and batch the processing of all points, allowing us to process all pixels in parallel to obtain the final result. The all pixel semantic features compose the whole semantic feature map, $S_f^{\text{all}} \in \mathbb{R}^{H \times W \times 16}$. The H is the image height and W is the image width. The channel number of feature maps is 16.

Finally, a decoder, one layer MLP is applied to generate semantic segmentation from the semantic feature maps

$$S^{\text{all}} = \text{Decoder}\left(S_f^{\text{all}}\right). \tag{10}$$

Replacing discrete class values with continuous semantic features offers two key advantages.

1) *Differentiability*: This transformation converts nondifferentiable problems into differentiable ones, enabling smoother optimization, albeit with a slight reduction in efficiency.
2) *Enhanced Expressiveness*: This approach increases the model's ability to represent semantic attributes in complex scenes, making it more effective at handling intricate segmentation tasks.

If the selected view has the ground truth, we set the cross-entropy loss to calculate semantic loss $\mathcal{L}_s^g$ for rendered semantic results S^{all} ground truth S^{GT}

$$\mathcal{L}_s^g = \text{CE}\left(S^{\text{all}}, S^{\text{GT}}\right). \tag{11}$$

It is worth noting that color splatting and semantic splatting are performed simultaneously and share the same spatial attributes, including $\mu^{3\text{-D}}$, $\Sigma^{3\text{-D}}$, S, and R. However, inadequate supervision remains a challenge in boundary regions under sparsely labeled views. To address this, we propose two strategies within the semantic splatting framework.

C. SAM for Pseudo-Labels

Challenges in scene segmentation often emerge in boundary regions, as these areas may extend beyond the common field of view of the supervisory views, leading to insufficient

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monitoring—particularly when only a limited number of supervisory views are available. While the Gaussian-like splatting architecture demonstrates notable rendering efficiency, its limited spatial continuity exacerbates the segmentation difficulty under our sparse label setting. To mitigate this challenge, we employ SAM2, a widely used zero-shot image/video segmentation method, to generate pseudo-labels for boundary regions. However, the results from SAM2 lack rich semantic information, making them unsuitable for direct use in semantic segmentation tasks.

We first randomly select a supervisory view and input the corresponding RGB image into SAM2 to generate a single-view instance segmentation result, treating regions that may correspond to objects as individual instances. Instances connected to the image boundary are defined as edge instances, allowing us to identify multiple edge instances and extract their corresponding masks. As a result, we generate a boundary mask B, where 0 represents nonboundary areas and 1 denotes boundary-intersecting areas. Next, we assign a pseudo-class to each instance based on the class with the maximum number of pixels within the corresponding instance area. Finally, we use each instance mask as a visual prompt and reinput it into SAM2 to generate multiview segmentation results for each instance, assigning the same pseudo-class label to all corresponding segmentation results. Through the above process, we can get the pseudo-label and boundary mask for each view, noted as S^p and B. If the selected view does not have ground truth, we define the loss $\mathcal{L}_s^p$ as follows:

$$\mathcal{L}_s^p = \mathrm{CE}\left(S^{\mathrm{all}} * B, S^p * B\right). \tag{12}$$

D. Aggregation Loss

The limited number of labeled views poses a significant challenge, as it hinders the optimization of semantic attributes for certain points in specific regions. Although the explicit representation used in our method provides notable advantages in optimization and rendering efficiency, its ability to continuously represent spatial information is weaker compared to implicit neural networks. This limitation affects the model's capacity to generalize effectively in unobserved areas. The primary cause of this limitation lies in the inherent differences in data structures. Implicit neural networks rely on MLPs to model the relationships between spatial locations and attributes, allowing them to be more robust to nearby spatial variations and enabling smooth transitions across unobserved regions. In contrast, our method utilizes an explicit point cloud structure, where spatial generalization is inherently constrained.

To address this challenge, we propose two levels of aggregation losses: one for the 2-D rendered feature map and the other for the 3-D spatial semantic features. The 2-D aggregation loss is designed to enhance the similarity of features from adjacent points in the rendered 2-D feature map, while the 3-D aggregation loss focuses on increasing the similarity of semantic features from neighboring points in 3-D space. Specifically, for the 2-D aggregation loss, we randomly select m feature points from the rendered feature map and calculate the mean similarity between each selected point and its k, e.g., $k = 5$ nearest neighbors, based on their 2-D pixel coordinates

$$\mathcal{L}_a^{\text{2-D}} = \frac{1}{mk} \sum_{i=1}^{m} \sum_{j=1}^{k} s_i \log\left(\frac{s_i}{s_j}\right). \tag{13}$$

For the 3-D aggregation loss $\mathcal{L}_a^{\text{3-D}}$, we adopt a similar approach. The key distinction lies in the selection process: m points are randomly chosen from the point cloud, and their 3-D coordinates are used to identify the k-nearest neighboring points.

E. Optimization Process, Details, and Total Loss

1) Optimization Process: For each optimization step, we select a view and obtain the rendered RGB and segmentation results. The captured images, along with their corresponding semantic labels or pseudo-labels, are used to optimize the parameters of the corresponding points in the point cloud. Given the inherent inaccuracies in pseudo-labels and their overwhelming proportion compared to ground-truth labels (e.g., 3%:97%), we manually set the ratio of views with ground-truth labels to pseudo-labeled views as 1:8. The ratio is important for the performance, the parameter experiments show the related results. We use COLMAP to extract the structure-from-motion (SfM) of the scene as the initial point cloud. The spatial structure of the point cloud is crucial for ensuring the accuracy of the rendered results, including both RGB images and segmentation outputs. Therefore, we employ split and merge strategies every 2000 steps for both the RGB attributes and semantic features of the points, adjusting them according to the gradient. This process is similar to villain Gaussian splatting.

2) Loss Function: We use $\mathcal{L}_1$ and $\mathcal{L}_{D-\mathrm{SSIM}}$ for optimizing color-related parameters. We define $\mathcal{L}_s^g$ as $\mathcal{L}_s$ when the selected view contains a ground truth label, and $\mathcal{L}_s^p$ as $\mathcal{L}_s$ when a selected view contains the pseudo-label. The $\mathcal{L}_s$ is used for optimizing semantic-related parameters. Specifically, if a view has ground truth, we do not use $\mathcal{L}_s^p$ for loss calculation. Instead, $\mathcal{L}_s^p$ is only applied to views without supervision, where we generate region masks and pseudo-labels for the boundary regions to compute the loss. For views with supervision, using $\mathcal{L}_s^g$ alone is sufficient. In addition, we employ $\mathcal{L}_a^{\text{2-D}}$ and $\mathcal{L}_a^{\text{3-D}}$ to increase the similarity between 2-D feature points and 3-D spatial points. The total loss is the weighted above losses

$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_{D-\mathrm{SSIM}} + \mathcal{L}_s + a * \mathcal{L}_a^{\text{2-D}} + b * \mathcal{L}_a^{\text{3-D}}. \tag{14}$$

Here, a and b are the hyper-coefficient for aggregation losses.

3) Details: We provide the parameter details of our method. The optimization process consists of 30 000 steps, with a maximum of 300 000 points in the point cloud. The number of semantic feature channels is set to 16, and the transfer layer maps these 16 channels to the number of semantic classes. For both $\mathcal{L}_a^{\text{2-D}}$ and $\mathcal{L}_a^{\text{3-D}}$, the number of neighboring points is set to $k = 5$. The candidate coefficient for $\mathcal{L}_a^{\text{2-D}}$ and $\mathcal{L}_a^{\text{3-D}}$ are 1.0, 0.5, and 0.1, respectively. The choice of coefficient for each subloss is crucial to performance, as demonstrated by our parameter experiments.

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TABLE I
DETAILS OF THE SYNTHETIC AND REAL DATASET

	#views	labeling ratio	#classes
sys #1	100	3%	20
sys #2	100	4%	18
sys #3	100	5%	20
sys #4	80	6%	19
sys #5	85	6%	19
sys #6	100	5%	18
real #1	100	2%	5
real #2	100	2%	3
real #3	100	2%	3

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset and Metrics

1) Dataset: In this article, we evaluate our method using the challenging dataset constructed in IRT [23]. This dataset comprises six synthetic subdatasets generated using the CARLA platform and three real subdatasets sourced from Google Maps. Each scene in the dataset is captured by extracting key frames from videos at equal intervals, resulting in approximately 100 images per scene taken from various shooting angles. The dataset includes a total of 865 images, each with a resolution of 512 × 512 pixels. As shown in Table I, only 2%–6% of the images in the training set are annotated with corresponding labels. The dataset covers scenes of varying scales, ranging from individual buildings to entire towns. Annotations for the synthetic data are generated by CARLA and include 20 types of ground objects such as buildings, roads, zebra crossings, and vegetation. For the real datasets, more than seven common ground object types are labeled, including buildings, roads, and vehicles. The images and corresponding annotations of the dataset are shown in Fig. 3.

2) Metric: In our experiment, we employ the mean intersection over union (mIoU), a widely used metric for segmentation tasks, to evaluate and compare the performance of different methods

$$\text{mIoU} = \frac{\text{TP}}{\text{FP} + \text{FN} + \text{TP}} \tag{15}$$

where TP is the true positive, FP is the false positive, and FN is the false negative.

B. Comparison Methods

To validate the effectiveness of our method, we selected several methods for comparative experiments. We compared our approach with training-based segmentation methods, including CNN-based techniques such as SegNet [67], Unet [68], DANet [36], and DeepLabv3 [69], as well as the Transformer-based SETR [16]. In addition, we primarily focused on comparisons with optimization-based methods. Sem-NeRF [22] performs joint color reconstruction and semantic segmentation by utilizing two MLP heads to process spatial features, predicting RGB values and semantic labels simultaneously. The baseline strategy employs a uniform renderer function for rendering both color results and semantic segmentation. Another two-stage strategy involves first reconstructing color information and then converting it into segmentation results [42]. IRT [23] enhances NeRF-based methods by incorporating CNN features as additional tokens to improve the processing of complex textures. The comparison methods are summarized as follows.

1) *SegNet [67]:* A CNN-based semantic segmentation method utilizing an encoder–decoder architecture.
2) *Unet [68]:* A CNN-based semantic segmentation method employing skip connections to preserve detailed information from encoder layers.
3) *DANet [36]:* A CNN-based semantic segmentation method introducing a dual attention mechanism to adaptively integrate local features and global dependencies.
4) *DeepLabv3 [69]:* A CNN-based semantic segmentation method leveraging dilated convolutions to effectively enlarge the receptive field.
5) *SETR [16]:* A Transformer-based semantic segmentation method.
6) *Sem-NeRF [22]:* The first NeRF-based model was designed for indoor image semantic segmentation.
7) *Color-NeRF [42]:* A NeRF-based method for semantic segmentation that introduces an additional color-radiance network to fuse pixel-level color information, improving segmentation performance.
8) *IRT [23]:* A NeRF-based method combining ray features with CNN features to enhance segmentation accuracy.

C. Overall Comparison Results

Quantitative Analysis: From Table II, it is evident that all optimization-based methods outperform training-based methods. This is primarily because training-based methods rely on learning the data distribution from large datasets to construct stable and generalized models. As a result, these methods struggle in tasks with limited data, particularly when only 3% of the input images are labeled. In contrast, optimization-based methods typically optimize a mapping from spatial point positions and associated attributes, including both color and semantic features, for each target scene. During the optimization process, the network can learn the 3-D information of the scene, ensuring view consistency in the results. When comparing Sem-NeRF, Color-NeRF, and IRT, the continuous integration of additional RGB information during semantic rendering improves segmentation accuracy. However, these methods face challenges related to high latency in both optimization and rendering, which limits their practical applicability.

Our method achieves higher mIoU scores than all the compared methods across all subdatasets, including both synthetic and real data. Specifically, compared to Sem-NeRF, our method outperforms it by an average of 24.42% in mIoU. We also surpass Color-NeRF by 3.54% on average mIoU. Notably, our method outperforms IRT, the current state-of-the-art method, by over 15.5% in average mIoU. Furthermore, our method exhibits significantly lower latency in both optimization and rendering compared to other optimization-based methods. For instance, our method saves 90% of testing time compared to IRT. This combination of high efficiency and accuracy makes our method more viable for practical applications.

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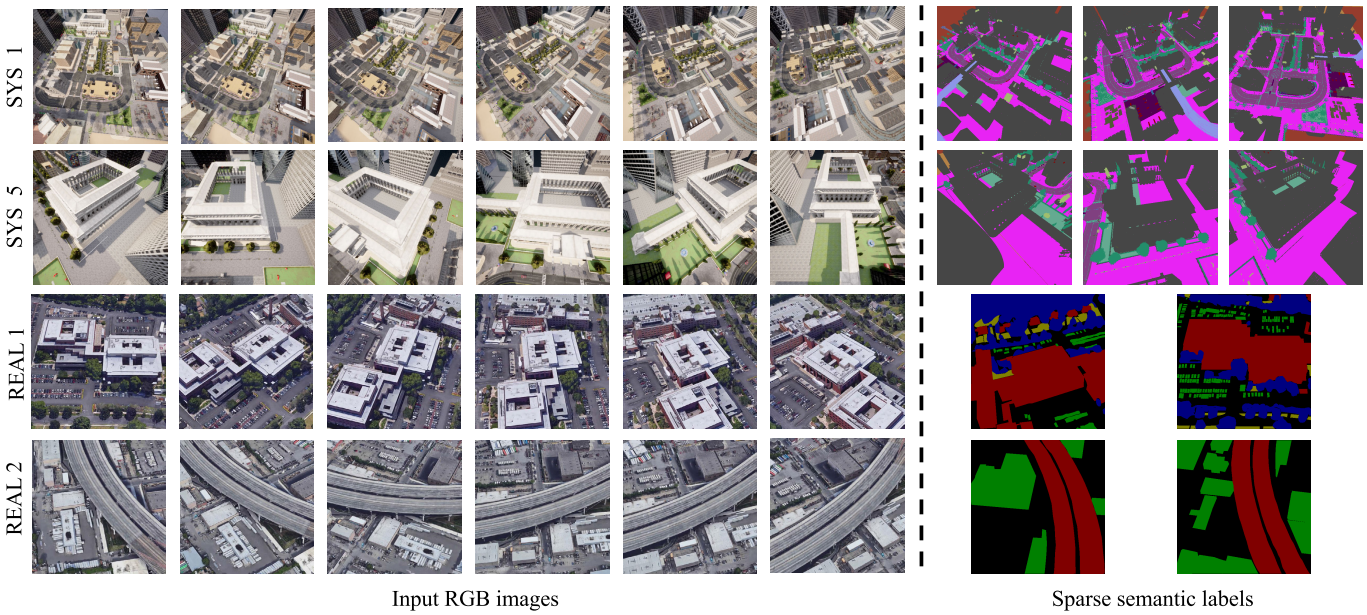


Fig. 3. We present input samples of four subdatasets. Left: six sampled input RGB images. Right: sparse semantic labels, with real scenes having only two labeled views. The type includes from area level (SYS 1 and REAL 1) to local building level (SYS5 and REAL 2).

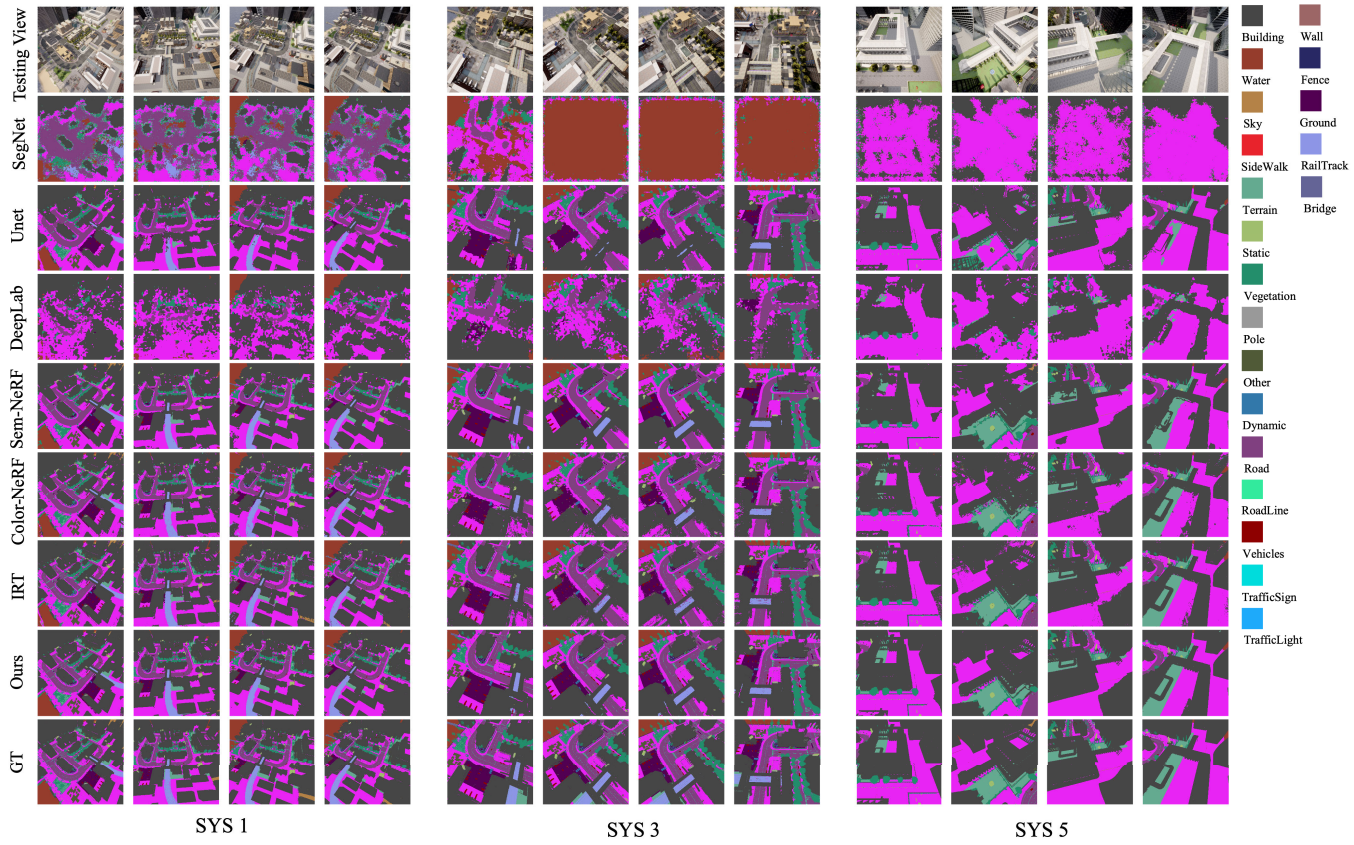


Fig. 4. Visual results from sys #1, sys #3, and sys #5. The results show that our method achieves more accuracy and outperforms other training-based and optimization-based methods in multiview segmentation takes for both complex synthesis and real scenes. We detail the class name of each color in the results.

We assessed the testing time of various methods. While training-based methods exhibit remarkable efficiency, their accuracy is insufficient for real-world applications. Conversely, optimization-based methods like Sem-NeRF, Color-NeRF, and IRT deliver higher accuracy but at the cost of extended computation times, requiring hours for optimization and several seconds to render results for each view. In contrast, our method achieves an excellent balance between accuracy

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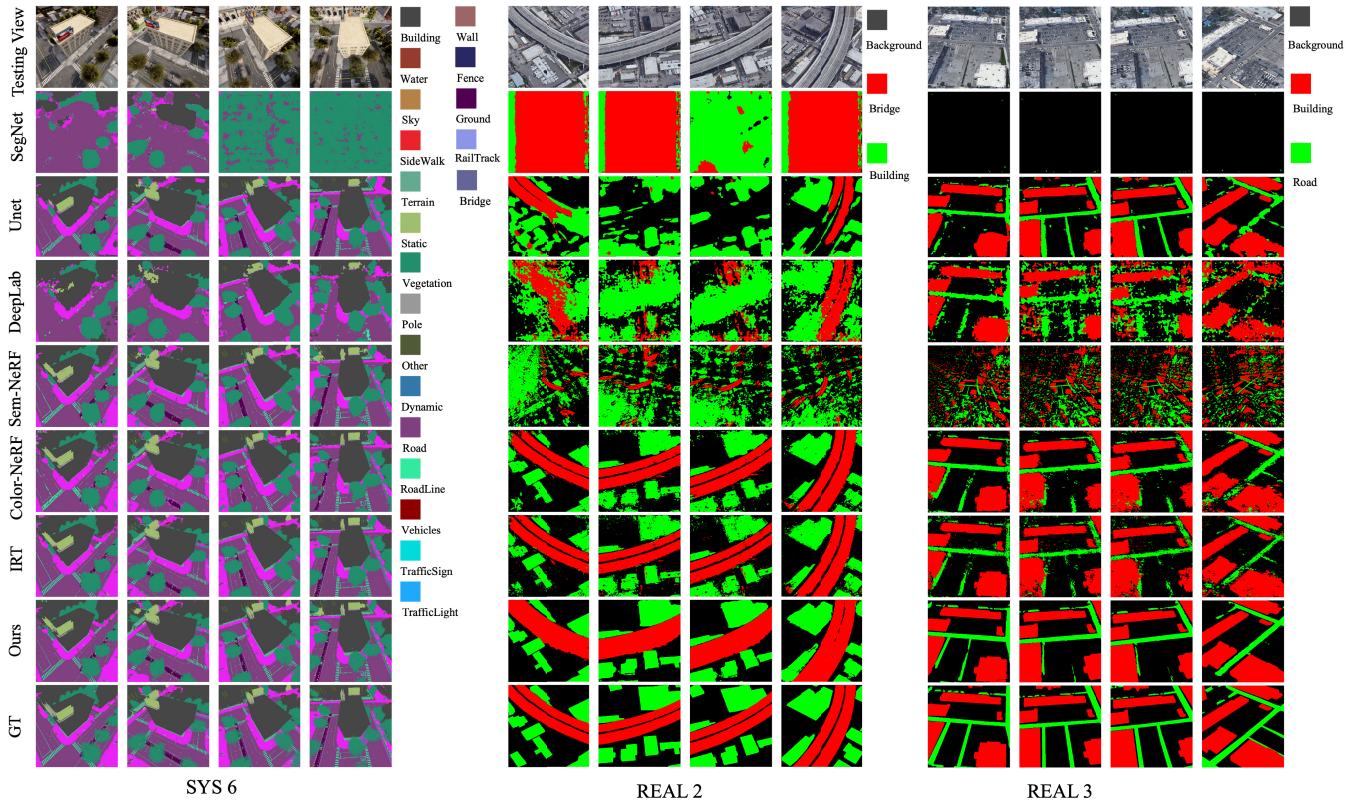


Fig. 5. Visual results from sys #6, real #2, and real #3. The results show that our method achieves more accuracy and outperforms other training-based and optimization-based methods in multiview segmentation takes for both complex synthesis and real scenes. We detail the class name of each color in the results.

TABLE II
MIoU METRIC AND TESTING TIME OF DIFFERENT METHODS ON EACH SUBDATASET

Method		sys #1	sys #2	sys #3	sys #4	sys #5	sys #6	real #1	real #2	real #3	AVG	Time
Training-based	SegNet	11.79	13.21	10.81	26.71	8.21	18.77	8.67	27.88	18.84	16.10	0.016s
	Unet	23.92	31.73	42.26	41.79	26.63	38.72	64.94	49.95	68.33	43.14	0.034s
	DANet	9.73	13.91	16.62	26.78	8.53	15.79	35.71	49.13	34.30	23.39	0.013s
	Deeplab	18.89	16.64	19.96	30.42	11.90	20.52	39.35	41.37	52.37	27.94	0.012s
	SETR	10.34	10.71	12.45	21.13	8.90	13.97	36.26	31.04	26.07	18.99	0.011s
Optimization-based	Sem-NeRF	55.73	34.81	49.82	57.24	41.44	41.85	11.64	18.78	19.76	36.79	10.743s
	Color-NeRF	58.03	38.46	50.86	59.14	43.77	43.53	62.04	85.85	69.92	56.84	19.552s
	IRT	57.86	**43.23**	53.47	59.98	43.73	48.59	**65.84**	84.31	71.02	58.67	50.662s
Optimization-based	Ours	**61.45**	38.98	**58.51**	**62.99**	**46.40**	**52.14**	63.31	**86.87**	**80.21**	**61.21**	**0.591s**

and efficiency. It combines high accuracy with drastically reduced latency, completing rendering in less than 1 s and requiring only about 10 min for optimization. This remarkable combination makes our approach significantly more practical and suitable for real-world applications.

D. Qualitative Analysis

1) Visually Analysis: We present results from six testing views across four selected synthetic subdatasets and two real subdatasets in Figs. 4 and 5. For each subdataset, we display the input RGB images alongside their corresponding semantic segmentation results. The comparison includes three training-based methods—SegNet, Unet, and DeepLab—and three optimization-based methods—Sem-NeRF, Color-NeRF, and IRT.

From the results, it is evident that optimization-based methods consistently outperform training-based methods, as discussed in the quantitative analysis section. Under sparse label conditions, training-based methods struggle to develop robust models capable of distinguishing complex background textures from object textures, often resulting in misclassification within background regions. For example, SegNet and DeepLab exhibit poor performance in background modeling for sys #1 and sys #6. In addition, training-based methods demonstrate significant limitations in segmenting small objects, as seen in the segmentation of small trees and zebra crossings in sys #1, as well as small roads in real #3, further highlighting their weaknesses in such scenarios. Furthermore, the visual results reveal poor view-consistency in training-based methods. This is primarily due to their inability to model spatial relationships effectively

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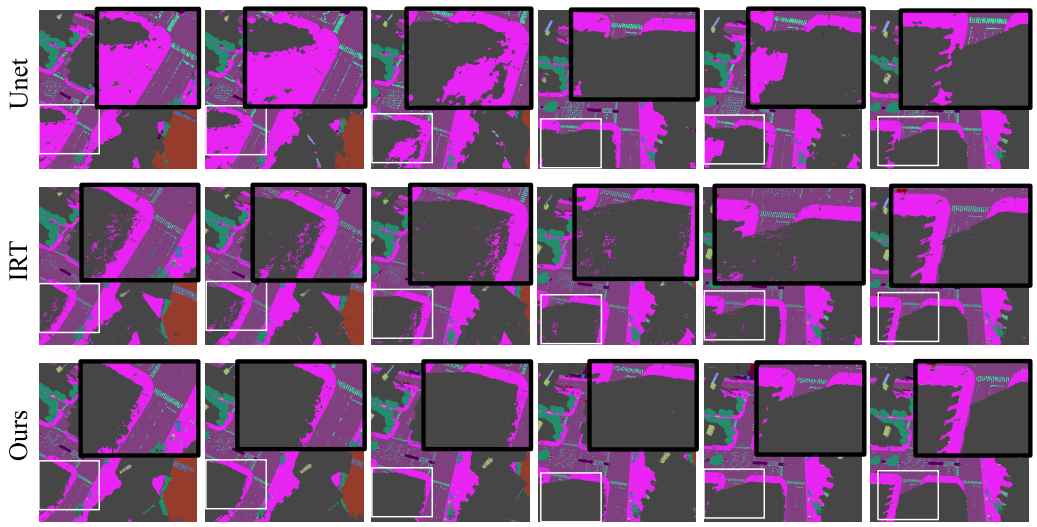


Fig. 6. View-consistent results from sys #4. The results show that our methods can generate results with more accuracy and view consistency. The labels in white boxes are not accurate or view-consistent using other methods. To provide a clearer comparison, we magnify the regions highlighted by white boxes and display them in the corresponding black boxes.

across different views, leading to inconsistent predictions and degraded performance in multiview segmentation tasks.

Optimization-based methods transform the segmentation task into a rendering process, producing class predictions for each pixel. This rendering approach offers distinct advantages: it achieves higher segmentation accuracy, even with sparse labels, and performs well in segmenting small objects. The visual results clearly demonstrate these benefits. Notably, our method delivers highly detailed and accurate segmentation results, as shown in roads in real#3 and building details in sys #5. However, other optimization-based methods, such as Sem-NeRF, Color-NeRF, and IRT, often incur significant latency during processing. In contrast, our approach leverages a splatting strategy to enhance efficiency. This remarkable combination of accuracy and efficiency positions our method as a promising solution for practical applications.

2) Semantic Consistency Comparison: We compare the segmentation view-consistency of different methods, including Unet, IRT, and our approach. As shown in Fig. 6, Unet demonstrates low accuracy in classifying buildings from the initial viewpoints. Since it processes images from each viewpoint independently, it fails to maintain view consistency. In contrast, IRT employs an implicit neural network to construct a continuous spatial mapping function, while our method utilizes a point cloud to represent the scene's spatial information, resulting in improved view consistency. This improvement can be attributed to the introduction of aggregation losses and pseudo-labels generated by SAM, which enhance the representation of boundary regions and preserve the structure of objects. For instance, our method produces more complete building structures. Compared to IRT, our approach not only ensures segmentation accuracy and view consistency but also achieves higher computational efficiency.

E. Ablation Study

In this section, we conduct ablation experiments to verify the following: 1) the necessity of pseudo-labels generated by SAM2; 2) the necessity of the 2-D aggregation loss $\mathcal{L}_a^{\text{2-D}}$; and 3) the necessity of the 3-D aggregation loss $\mathcal{L}_a^{\text{3-D}}$. We use the method that employs only basic semantic feature splatting as our baseline. The evaluation results are presented in the Table III.

First, we incorporate pseudo-labels generated by SAM2 into the optimization process. Due to the weaker spatial continuity of the point cloud's explicit spatial structure compared to implicit neural networks, our proposed method initially exhibited suboptimal performance in boundary regions. To address this, we use SAM2 to generate view-consistent pseudo-labels as additional supervisions specifically for these regions. As shown in Table III, this strategy proves effective across almost all subdatasets. With pseudo-labels, the method outperforms baseline over 0.4%. Fig. 7 also presents the visual results, clearly demonstrating the improved effectiveness in boundary regions, particularly in the buildings within the sys #1 subdataset, better to zoomed in.

Second, we conduct experiments to validate the necessity of the two aggregation losses. We gradually incorporate $\mathcal{L}_a^{\text{2-D}}$ and $\mathcal{L}_a^{\text{3-D}}$ into the optimization process. The $\mathcal{L}_a^{\text{2-D}}$ loss encourages semantic features within the same class to become more similar, while $\mathcal{L}_a^{\text{3-D}}$ ensures that semantic features in neighboring spatial regions are also similar. Notably, due to the splatting strategy, $\mathcal{L}_a^{\text{3-D}}$ indirectly influences the semantic feature map. As shown in Table III, incorporating $\mathcal{L}_a^{\text{2-D}}$ improves performance by 1.1%, and further adding $\mathcal{L}_a^{\text{3-D}}$ increases performance by an additional 2.4%, demonstrating the effectiveness of these losses in enhancing segmentation quality. We apply principal component analysis (PCA) to reduce the dimensions of the splatted semantic features from 16 to 3, enabling the generation of RGB visual representations, as shown in Fig. 8. From the results on the sys #6 subdataset, we observe a significant improvement in the similarity of features within the same class and the consistency of features across different viewing angles.

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TABLE III
ABLATION STUDY RESULTS

Baseline	+Pseudo Labels	+ $\mathcal{L}_a^{2D}$	+$\mathcal{L}_a^{2D}$	sys #1	sys #2	sys #3	sys #4	sys #5	sys #6	real #1	real #2	real #3	AVG
✓				58.51	29.11	56.96	60.62	41.35	50.18	58.96	84.91	76.46	57.45
✓	✓			59.54	26.81	56.98	59.62	44.05	49.01	61.71	83.13	79.62	57.82
✓	✓	✓		60.27	29.08	58.15	60.28	45.51	49.63	62.03	84.33	80.11	58.82
✓	✓	✓	✓	**61.45**	**38.98**	**58.51**	**62.99**	**46.40**	**52.14**	**63.31**	**86.87**	**80.21**	**61.21**

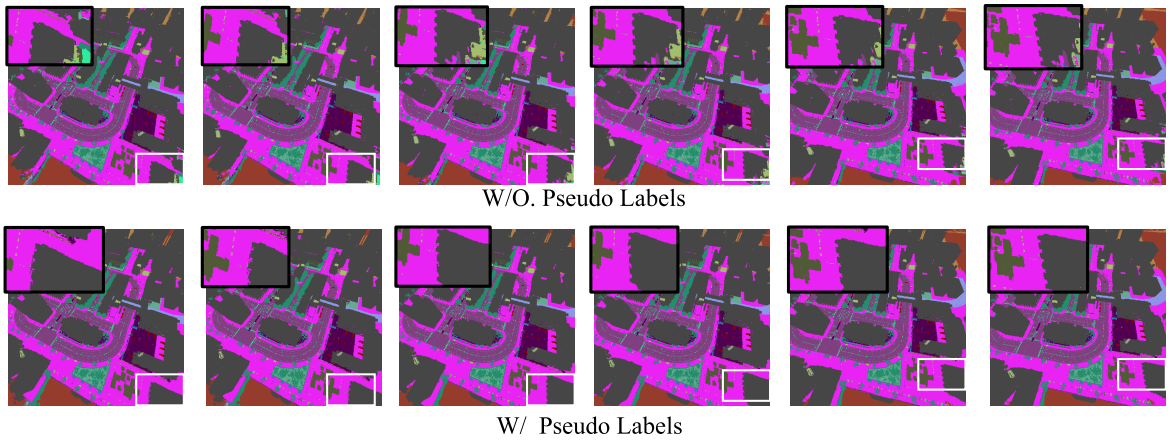


Fig. 7. Pseudo-labels for boundary regions assist the network in improving accuracy in these challenging areas (shown in white boxes).

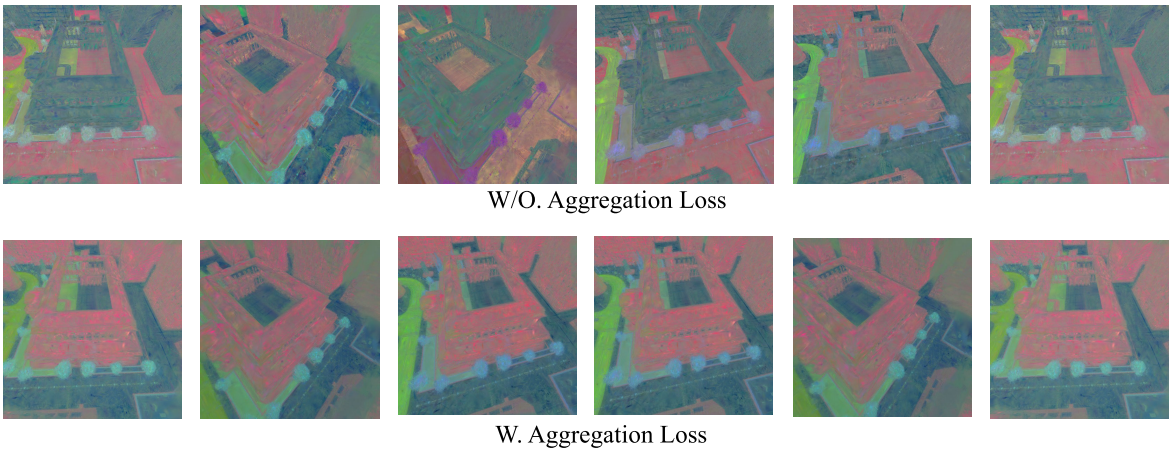


Fig. 8. We calculate the PCA results of a splatted semantic feature map. The results show that aggregation losses help the network enhance the stability of semantic features.

F. Parameter Study

We observe that the hyper-parameters, including the ratio of two types of views (with ground truth and with pseudo-label), the weights of $\mathcal{L}_a^{2\text{-D}}$ and $\mathcal{L}_a^{3\text{-D}}$ and the number of adjust point in aggregation losses, play a crucial role in the optimization process of our method. To determine the optimal values for these parameters, we conduct experiments on both two synthetic subdataset (sys #1 and #5) and two real subdataset (real #1 and #3).

1) Ratio of Two Types Views: The pseudo-labels generated by SAM2 contain some inaccuracies and are not sufficiently precise for semantic segmentation. Furthermore, the proportion of pseudo-labeled data significantly outweighs the ground truth data, e.g., 97% pseudo-labels versus 3% ground truth. Selecting views with ground truth and pseudo-labels with equal probability would lead to imbalanced supervision, resulting in suboptimal performance. To mitigate the negative impact of pseudo-labels, we experimented with different ratios of selected two types of views. An insufficient proportion of pseudo-labels results in inadequate supervision of boundary areas, while an excessive proportion introduces inaccuracies and unrefined labels, negatively impacting the final results. As shown in Table IV, our method achieves the best performance on both SYS #1, SYS #5, REAL #1, and REAL #3 when the ratio of views with ground truth to views with pseudo-labels is set to 1:8.

2) Weight of $\mathcal{L}_a^{2\text{-D}}$: The aggregation loss $\mathcal{L}_a^{2\text{-D}}$ is pivotal in enhancing feature similarity within the same class at the level of the 2-D semantic feature map. By enforcing semantic

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TABLE IV
RESULTS OF DIFFERENT RATIOS OF TWO TYPES VIEWS

Ratio	sys #1	sys #5	real #1	real # 3	AVG
1:7	59.10	43.85	61.32	78.99	60.79
1:8	**59.54**	**44.05**	**61.71**	**79.62**	**61.23**
1:9	59.46	43.97	61.47	79.05	60.99

TABLE V
RESULTS OF DIFFERENT COEFFICIENTS FOR $\mathcal{L}_a^{2\text{-D}}$

Weight	sys #1	sys #5	real #1	real #3	AVG
1.0	60.10	45.03	**62.17**	79.23	61.63
0.5	**60.27**	**45.51**	62.03	**80.11**	**61.98**
0.1	58.81	45.24	60.58	80.03	61.17

TABLE VI
RESULTS OF DIFFERENT COEFFICIENTS $\mathcal{L}_a^{3\text{-D}}$

Weight	sys #1	sys #5	real #1	real #3	AVG
1.0	60.38	45.65	57.44	79.01	58.91
0.5	61.05	46.01	59.94	79.93	60.50
0.1	**61.45**	**46.40**	**63.31**	**80.21**	**62.38**

TABLE VII
RESULTS OF DIFFERENT k

K	sys #1	sys #5	real #1	real #3	AVG
3	60.33	45.26	62.45	78.84	61.72
5	**61.45**	**46.40**	**63.31**	**80.21**	**62.84**
8	61.20	45.97	62.37	78.99	62.13

similarity among neighboring feature points, it promotes a more compact and cohesive feature distribution for points belonging to the same class. This not only boosts the segmentation model's discriminative power but also ensures improved semantic continuity across segmented regions, especially at class boundaries. Such continuity effectively reduces misclassification errors and boundary inconsistencies, leading to cleaner and more accurate segmentation results. Experimental results, as presented in Table V, confirm that carefully tuning the weight coefficient $\mathcal{L}_a^{2\text{-D}}$ yields the best tradeoff between feature consistency and segmentation performance. These findings underscore the critical role of optimizing the $\mathcal{L}_a^{2\text{-D}}$ coefficient to achieve superior overall performance. The best coefficient of $\mathcal{L}_a^{2\text{-D}}$ is 0.5.

3) Weight of $\mathcal{L}_a^{3\text{-D}}$: Similar to $\mathcal{L}_a^{2\text{-D}}$, the aggregation loss $\mathcal{L}_a^{3\text{-D}}$ is designed to enhance feature similarity, but at the 3-D spatial point level. By enforcing semantic similarity among spatially adjacent points in the 3-D point cloud, $\mathcal{L}_a^{3\text{-D}}$ ensures that points belonging to the same class exhibit consistent features across the spatial domain. This improves spatial coherence in segmentation and strengthens the alignment between the 3-D structure of the point cloud and its semantic representation. The weight coefficient $\mathcal{L}_a^{3\text{-D}}$ controls the relative impact of $\mathcal{L}_a^{3\text{-D}}$ during the optimization process. As shown in Table VI, experiments evaluating different values of $\mathcal{L}_a^{3\text{-D}}$ reveal its significant influence on segmentation quality. While $\mathcal{L}_a^{2\text{-D}}$ focuses on local feature similarity within 2-D feature maps, $\mathcal{L}_a^{3\text{-D}}$ extends this consistency to the 3-D spatial level. Their combined application ensures both precise local segmentation in 2-D and robust spatial consistency in 3-D. The results in Table III demonstrate that leveraging these two losses together leads to substantial improvements in segmentation accuracy and multiview consistency. The choice $\mathcal{L}_a^{2\text{-D}}$ is 0.1.

4) Choice of the K: In the aggregation loss, we compute the similarity of semantic features among K neighboring points at both the 3-D spatial level and the 2-D feature map level. To analyze the impact of different K values, we conducted experiments on sys #1, sys #5, real #1, and real #3. Our results indicate that using too few neighboring points fails to effectively mitigate semantic feature inconsistencies, while an excessively large number of neighbors leads to feature oversmoothing. The results are summarized in Table VII.

V. FURTHER WORK AND CONCLUSION

In this article, our method demonstrates the efficiency and accuracy of splatting technology in multiview image segmentation for remote sensing scenes. Moving forward, we plan to explore more appropriate 3-D structure representations for sparse spatial characteristics, such as using linear functions instead of Gaussian functions. In addition, we aim to integrate implicit neural functions with splatting technology to estimate a semantic neural function for each point in the point cloud, further improving spatial consistency.

In conclusion, we aim to improve multiview image segmentation for remote sensing of target scenes, especially under conditions with sparse semantic labels. We propose an efficient semantic splatting method based on Gaussian splatting to address this challenge. To overcome the lack of labels in boundary regions, we introduce SAM2, which generates pseudo-labels for these areas. Furthermore, we design two aggregation losses: one at the 2-D feature map level and another at the 3-D spatial level, to enhance the spatial continuity of our method and boost its performance. Experimental results demonstrate that our approach achieves significant improvements in segmentation accuracy and low latency, providing a solid foundation for practical applications.

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